

Numerical Solution and Error Analysis of the Newtonian Two-Body Problem

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Abstract

In this work, we compute the numerical solution of the Newtonian two-body problem and compare it to the analytical Kepler solution. Using Mercury's orbital parameters, we solve the radial and angular differential equations, construct the numerical trajectory, derive the analytical orbit, and quantify the numerical errors in conserved quantities and in position.

1 Introduction

The Newtonian two-body problem describes the motion of two spherically symmetrical point masses interacting under gravity, the equation of motion is

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + \frac{1}{r}\frac{d}{dt}(r^2\dot{\theta})\hat{\boldsymbol{\theta}}, \quad (1)$$

where the second term vanishes in the gravitational case because angular momentum is conserved:

$$l = r^2\dot{\theta} = \text{constant}. \quad (2)$$

The radial equation then becomes:

$$\ddot{r} = \frac{l^2}{r^3} - \frac{k}{r^2}. \quad (3)$$

The analytical solution is a Kepler ellipse:

$$r(\theta) = \frac{p}{1 - e \cos \theta}, \quad p = \frac{l^2}{k}. \quad (4)$$

The goal of this work is to (1) solve Eqs. (3)–(4) numerically, (2) compare the resulting Cartesian trajectory with the analytical ellipse, and (3) compute errors in angular momentum conservation and in the position vector.

2 Numerical Solution

Mercury's aphelion and perihelion distances were used to compute the initial tangential velocity and specific angular momentum. The radial equation was integrated over one orbital period. The angular coordinate was obtained by integrating $\dot{\theta}(t) = \frac{l}{r(t)^2}$. Cartesian coordinates were reconstructed via $x = r \cos \theta$, $y = r \sin \theta$.

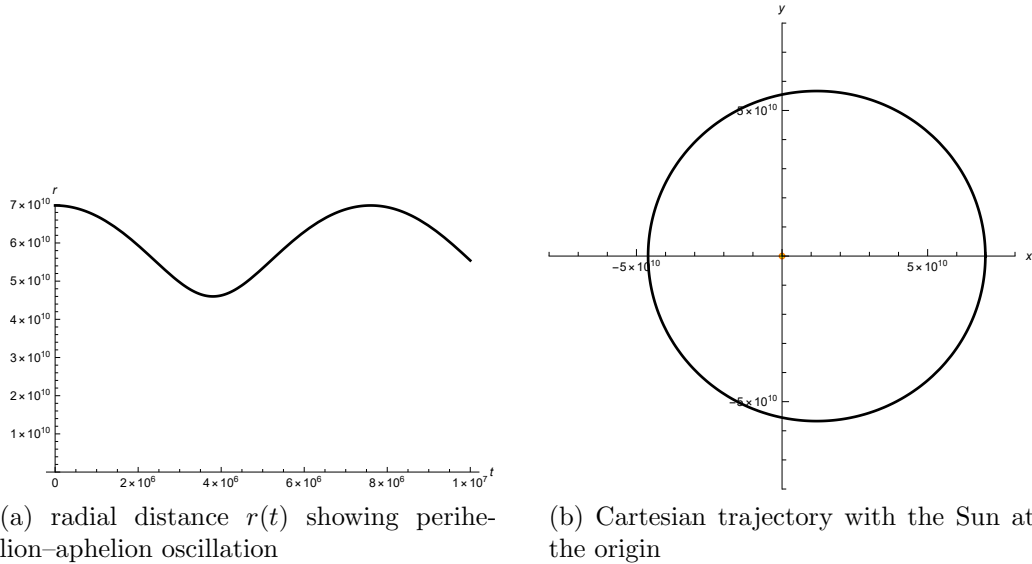


Figure 1: Classical numerical solution of Mercury's orbit.

2.1 Symplectic Method

In addition to the classical numerical integration of the radial equation, we implemented a symplectic leapfrog method to solve the reduced Hamiltonian form of the two-body problem. The Hamiltonian is

$$H(r, p_r, p_\theta) = \frac{p_r^2}{2} + \frac{p_\theta^2}{2r^2} - \frac{k}{r}. \quad (5)$$

Since $p_\theta = \ell$ is conserved, the system reduces to

$$H(r, p_r) = \frac{p_r^2}{2} + \frac{\ell^2}{2r^2} - \frac{k}{r}. \quad (6)$$

We employ the leapfrog (velocity-Verlet) scheme:

$$p_r \left(t + \frac{\Delta t}{2} \right) = p_r(t) - \frac{\Delta t}{2} \frac{\partial H}{\partial r}, \quad (7)$$

$$r(t + \Delta t) = r(t) + \Delta t p_r \left(t + \frac{\Delta t}{2} \right), \quad (8)$$

$$p_r(t + \Delta t) = p_r \left(t + \frac{\Delta t}{2} \right) - \frac{\Delta t}{2} \frac{\partial H}{\partial r}. \quad (9)$$

This method is symplectic, time-reversible, and preserves energy with bounded oscillations instead of secular drift. The resulting trajectory is obtained by simultaneously evolving θ using $\dot{\theta} = \frac{\ell}{r^2}$.

3 Analytical Kepler Orbit

Using initial conditions, the specific orbital energy and angular momentum were computed. These yield e , a , $p = a(1 - e^2)$, which define the closed-form Kepler solution.

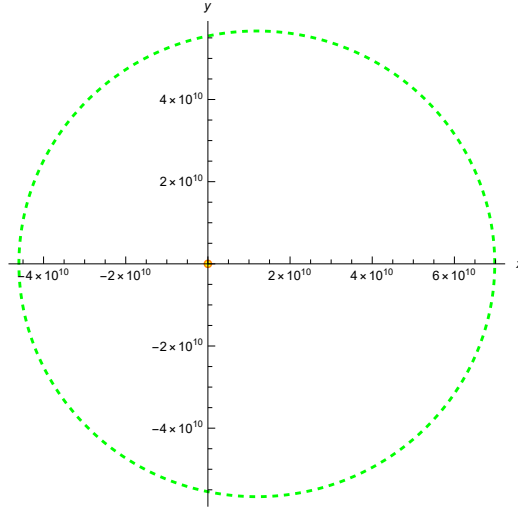


Figure 2: Analytical Keplerian trajectory plotted using the closed-form expression for $r(\theta)$.

A combined plot of the analytical, classical numerical, and symplectic trajectories confirms close agreement.

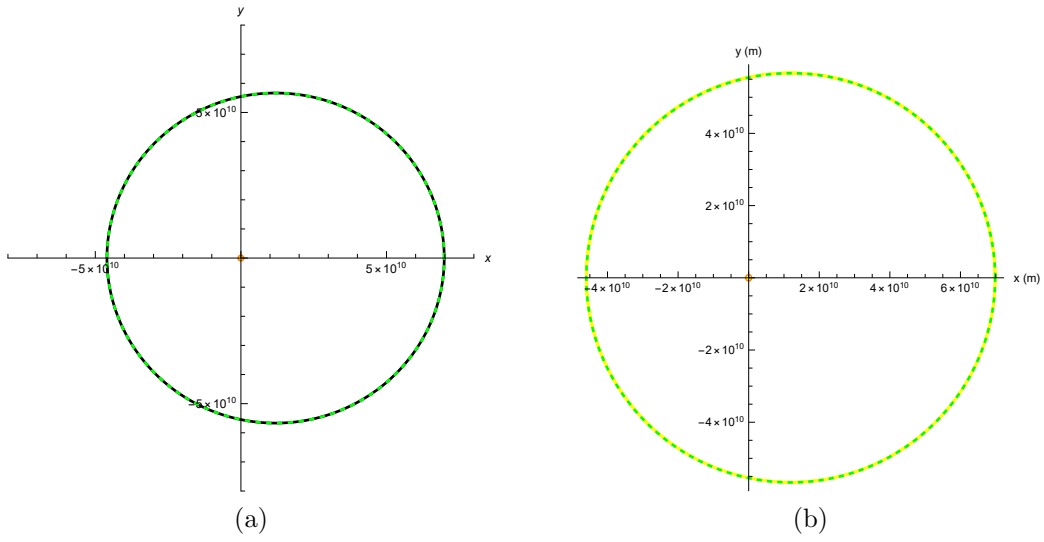


Figure 3: Superposition of analytical, classical numerical, and symplectic trajectories.

4 Error Analysis

4.1 Energy and Angular Momentum Conservation

The relative change in specific energy, $\Delta E/E_0$, was computed at 100 sample times. Similarly, $\frac{\ell(t)-\ell_0}{\ell_0}$ was computed.

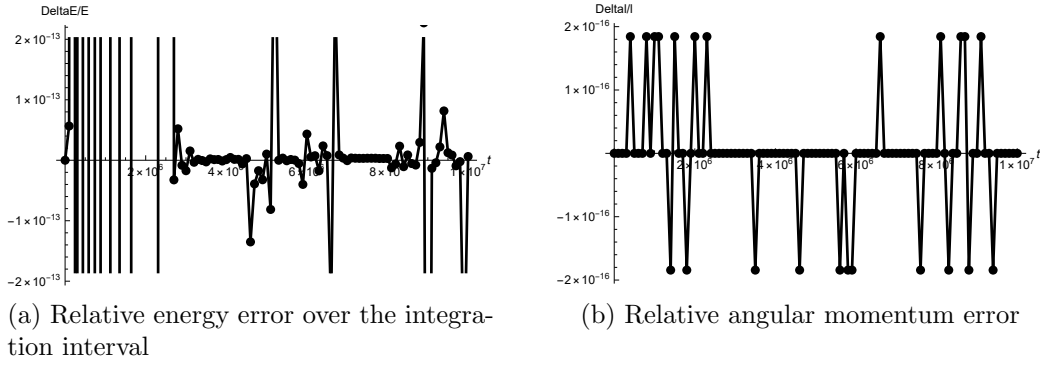


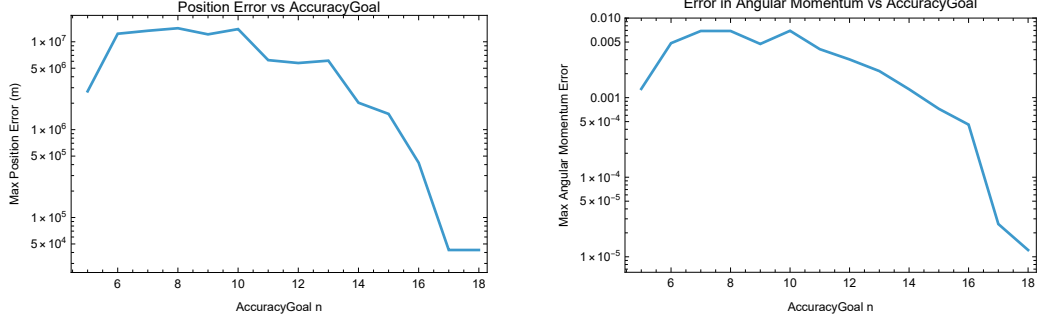
Figure 4: Comparison of conserved-quantity errors in the numerical solution

4.2 Position Error

The deviation between the numerical and analytical trajectories was computed using the scalar radial error

$$\Delta r(t) = \sqrt{(x_{\text{num}}(t) - x_{\text{an}}(t))^2 + (y_{\text{num}}(t) - y_{\text{an}}(t))^2}.$$

This quantity measures the instantaneous distance between the two trajectories in Cartesian space.



(a) Radial position error $\Delta r(t)$ vs. accuracy goal

(b) Angular momentum error vs. accuracy goal

Figure 5: Comparison of numerical error metrics: (a) radial position error $\Delta r(t)$ and (b) angular momentum error, shown as functions of the solver accuracy goal.

4.3 Orbital Period Comparison

Method	Period	Relative Error
Analytical	$7.600718332601069 \times 10^6$	0
Symplectic (manual)	$7.600713872578831 \times 10^6$	$-5.867895695969338 \times 10^{-7}$
Symplectic (Leapfrog)	$7.600714131145510 \times 10^6$	$-5.527708534548106 \times 10^{-7}$
Runge-Kutta	$7.600714131148754 \times 10^6$	$-5.527704265572848 \times 10^{-7}$

Figure 6: Comparison of Mercury orbital periods obtained from analytical, Runge–Kutta, and symplectic methods. Relative errors are computed with respect to the analytical period.

This table summarizes the orbital periods obtained using different methods and their relative errors with respect to the analytical Keplerian period T_{an} . The analytical solution yields $T_{\text{an}} = 7.600718332601069 \times 10^6$, which serves as the reference value and therefore has zero relative error.

The manually implemented symplectic integrator produces a period that differs from the analytical value by a relative error of order 10^{-7} . This discrepancy is primarily attributed to the use of a fixed step size and the ab-

sence of adaptive error control in the manual implementation. This error was achieved by taking number of steps as 10000.

The symplectic leapfrog method implemented via Mathematica's built-in symplectic partitioned Runge–Kutta scheme achieves improved agreement with the analytical period, exhibiting a relative error of approximately -5.53×10^{-7} . This confirms the expected advantage of symplectic methods in preserving the Hamiltonian structure and limiting long-term drift in orbital quantities.

The classical Runge-Kutta method yields a period nearly identical to that obtained with the built-in symplectic method, with a comparable relative error. This indicates that, for the chosen tolerances and integration interval, both methods achieve similar accuracy in determining the orbital period, despite their different long-term stability properties. The RK method is the leading method in performance; the speed of execution was remarkable, there was no need for any relaxation constraints on this method, the error is the smallest amongst the others and it was the simplest to be coded.

Overall, all numerical methods reproduce the analytical Kepler period with relative errors on the order of 10^{-7} , demonstrating excellent agreement for this ideal Newtonian two-body problem. Any remaining discrepancy arises from numerical discretization, interpolation of turning points, and finite step-size effects rather than from physical modeling assumptions.

5 Conclusions

We have compared three solutions of the Newtonian two-body problem. The analytical trajectory serves as the exact reference. The classical numerical solution is accurate but exhibits small energy drift and slight differences in period determination. The symplectic Hamiltonian integrator would've preserve energy and geometric structure exceptionally well, yielding an orbit nearly indistinguishable from the analytical one and an improved agreement in orbital period, but given the hand calculations, the errors were slightly higher than expected, so it needs to be enhanced further more.

References

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